

LESSON 6-4 • HOMEWORK PACKET • SHEET 1 OF 2

Writing Numerical Expressions

Maryland Standard 6.EE.A.2 • Learning target – I can write a numerical expression that matches a situation.

REMEMBER

- A **numerical expression** uses numbers and operations. It has no equals sign.
- Sum (+), difference (–), product ($\times$), quotient ($\div$).
Less than flips the order: 5 less than 12 is $12 - 5$.
- Use **parentheses** for a part done first: the sum of 4 and 6, tripled, is $3 \times (4 + 6)$.

WORKED EXAMPLE

Kayla buys 3 notebooks at \$4 each and a backpack for \$15. Write an expression for the total cost. Then evaluate it.

Step 1 Notebooks: 3×4 . Backpack: 15

Step 2 Join them: $3 \times 4 + 15$

Step 3 Evaluate: $12 + 15 = 27$

Answer $3 \times 4 + 15 = \$27$

PRACTICE

show your work • label units

1. Write a numerical expression for each phrase. Do not evaluate.

- a. the product of 7 and 8, minus 10 _____ b. 12 less than 50 _____
- c. the quotient of 36 and 9, plus 2 _____ d. the difference of 15 and 6, doubled _____
- e. 9 more than the product of 3 and 5 _____ f. the quotient of 42 and 7, tripled _____

2. **Field Day.** A class buys 5 packs of water at \$6 each and 4 bags of oranges at \$3 each. Write one expression for the total cost. Then evaluate it.

Show your work.

Expression and cost _____

3. **Bake Sale.** Nia sells 12 muffins at \$2 each and 8 loaves of bread at \$5 each. She spent \$15 on supplies. Write one expression for the money she has left after paying for supplies. Then evaluate it.

Show your work.

Expression and amount _____

4. Which expression matches “the sum of 8 and 2, times 5”?

- ☐ $8 + 2 \times 5$ ☐ $(8 + 2) \times 5$ ☐ $8 \times 5 + 2$

5. **Write a Story.** Write a short story that matches $4 \times (6 + 3)$. Then evaluate the expression.

CHECK YOURSELF

- ☐ I can do this on my own ☐ I need a little help ☐ I have a question about # _____

AT HOME Find two prices at home or in an ad. Write one expression for buying some of each.

LESSON 6-4 • HOMEWORK PACKET • SHEET 2 OF 2**Order of Operations with Powers**

Maryland Standard 6.EE.A.2c • Learning target – I can evaluate an expression that contains grouping symbols and powers using the order of operations.

REMEMBER

- Order: **1.** grouping () **2.** powers **3.** $\times$ and $\div$ left to right **4.** $+$ and $-$ left to right.
- $\times$ and $\div$ are partners: do whichever comes first from the left. So are $+$ and $-$.
- A power applies only to the number just before it: $2 \times 3^2 = 2 \times 9 = 18$.

WORKED EXAMPLE

Evaluate $20 - (2 + 1)^2 \times 2$.

Step 1 Grouping, then power: $20 - 3^2 \times 2 = 20 - 9 \times 2$

Step 2 Multiply, then subtract: $20 - 18 = 2$

Answer 2

PRACTICE

show your work • label units

- 1.** Evaluate each expression.

a. $6 + 4 \times 5 = \underline{\hspace{2cm}}$

b. $(6 + 4) \times 5 = \underline{\hspace{2cm}}$

c. $3^2 + 2 \times 4 = \underline{\hspace{2cm}}$

d. $48 \div (2 + 6) = \underline{\hspace{2cm}}$

e. $5^2 - 10 \div 2 = \underline{\hspace{2cm}}$

f. $36 \div 3 \times 2 = \underline{\hspace{2cm}}$

- 2.** Add one pair of parentheses to make each statement true.

a. $2 + 3 \times 4 = 20$

b. $24 \div 4 + 2 = 4$

- 3. Tile Floor.** A square floor is 8 tiles by 8 tiles. Mr. Ruiz replaces all but 4 tiles. Each new tile costs \$3. Evaluate $3 \times (8^2 - 4)$ to find the cost.

Show your work.

Answer _____ dollars

- 4. Bead Kits.** Ms. Lewis has 5 bags with 4^2 beads in each bag. Her class uses 30 beads. Evaluate $5 \times 4^2 - 30$ to find how many beads are left.

Answer _____ beads

- 5. Who Is Right?** For $18 - 6 + 2$, Kim got 10 and Lou got 14. Who is correct?

☐ Kim

☐ Lou

TEST PRACTICE

choose the best answer

- 6.** What is the value of $5 + 3 \times 2^2$?

A 17

B 32

C 41

D 256

- 7.** Which expression has a value of 14?

A $(2 + 3) \times 4$

B $2 + 3 \times 4$

C $2^3 + 4$

D $16 \div 2 \times 4$

CHECK YOURSELF

☐ I can do this on my own

☐ I need a little help

☐ I have a question about #_____

AT HOME Show someone at home why $2 + 3 \times 4$ and $(2 + 3) \times 4$ have different values.